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EDEXCEL - LONDON EXAMINATIONS

Stewart House 32 Russell Square London WC1B 5DN

FINAL

January 2001

HMK

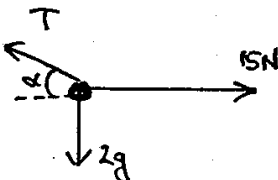
Advanced Supplementary/Advanced Level

17-01-01

General Certificate of Education

Subject MECHANICS 6677

Paper No. M1

Question number	Scheme	Marks
1. (a)	Resolving vertically e.g. $R_p + R_q = 70$ $R_p = 20 \Rightarrow R_q = 50$	M1 A1 (2)
(b)	A valid moments equation e.g. $R_p \times 0.5 + R_q \times x = 70 \times \frac{3}{2}$ $20 \times 0.5 + 50 \times x = 70 \times \frac{3}{2}$ Completing method to find AQ $AQ = 1.9$	M1 A1 ft DM1 A1 cao (4)
2	 (a) <u>ONE</u> resolution equation e.g. $T \cos \alpha = 15$ or $T \sin \alpha = 2g$ are most likely but $T = 15 \sec \alpha + 2g \sin \alpha$, $2g \cos \alpha = 15 \sin \alpha$ also possible as is also ham's chosen. One equation correct; second independent eqn. correct (omission of g loses A1 only) $\tan \alpha = \frac{2g}{15}$ or $\frac{2}{15}$ [$\tan \alpha = \frac{15}{2g}$ scores M1 A0] Answer for α as 53° or 52.6 (b) Using <u>valid</u> equation (line 1 M1 required) to extract value of T (or eliminating α from valid eqns) $T = 24.7$ or 25	M1 A1 + A1 M1 A1 ft A1 (6) M1 A1 (2)

("Over accurate" answers in (a) or (b) or both which round to correct) answer receive a penalty of -1 once overall

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3.(a)	For particle A $T - 3mg = 3ma$ [Note $T - mg = ma$ or $T - m = ma$ etc score M1] $T - 3mg = 3m(\frac{2}{5}g) \rightarrow T = \frac{21}{5}mg$	M1 A1 $\rightarrow$ A1(3)
(b)	String is inextensible	B1 (1)
(c)	For particle B $kmg - T = kma$ (or system) $kmg - 3mg = (km + 3m)a$ $kg - \frac{21}{5}g = \frac{2}{5}kg$ (or equivalent equation in k only) Solving DM1 dependent on first M1 in (c) $k = 7$	M1 A1 f.t. DM1 A1 cas (4)
(d)	Tension is of same magnitude throughout the string	B1 (1)
4.(a)	At $t=0$ $\underline{r}_P = 2\underline{i} - \underline{j}$; At $t=2$, $\underline{r}_P = 6\underline{i} + \underline{j}$ Velocity of P constant $\Rightarrow \underline{v}_P = \frac{(6\underline{i} + \underline{j}) - (2\underline{i} - \underline{j})}{2}$ $\underline{v}_P = 2\underline{i} + \underline{j}$ (one slip in $\underline{i}$ or $\underline{j}$ only)	M1A1 A1 f.t. (3)
(b)	$\arctan \frac{1}{2}$ (or $\arctan 2$ allowed for M1) 26.6° only	M1 A1 (2)
(c)	$\vec{OC} = 2\underline{i} - \underline{j} + 5(2\underline{i} + \underline{j})$ OR $6\underline{i} + \underline{j} + 3(2\underline{i} + \underline{j})$ $\vec{OC} = 12\underline{i} + 4\underline{j}$ $ \vec{OC} = \sqrt{12^2 + 4^2}$ $OC = 12.6$ only or equivalent f.t. answer given to 1 decimal place also depends on M1+M1	M1 A1 f.t. M1 A1 f.t. (4)

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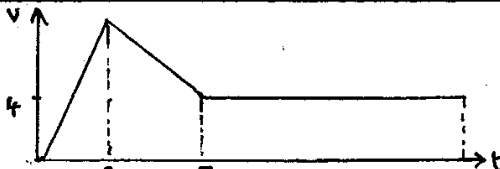
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Paper No. M1

Question number	Scheme	Marks
5.(a)	Before $\rightarrow 4.5$ $\leftarrow 3$ Momentum conserved on system $A \text{ (0.6)} \quad B \text{ (0.2)}$ After $\frac{v}{2} \rightarrow v \rightarrow$ $0.6 \times 4.5 - 0.2 \times 3 = 0.6 \times \frac{v}{2} + 0.2 \times v$ <u>Solving for v</u> $\rightarrow v = 4.2$	M1 A1 <u>M1</u> A1 (4)
(b)	Change in momentum of A or B attempted $0.2(3 + 4.2)$ OR $0.6(4.5 - 2.1) \rightarrow 1.44$ units <u>NS</u>	M1 <u>A1</u> ft <u>B1</u> (3)
(c)	$R = mg$ $\mu R = \mu mg = \text{retarding force or deceleration } \mu g$ $v^2 = u^2 + 2as$ applied $0 = 4.2^2 - 2\mu g \times 2$ (or equivalent work) $\mu = \frac{4.2^2}{4g} = 0.45$ (DM1 depends on M1+1)	B1 M1 M1 A1 ft DM1 A1 (6)
6 (a)	 G1 2 stages ✓ slope G2 3 stages ✓ slope + G1 for 2, 7, 4 on axes	G3, 2, 1, 0 (3)
(4)	Using $v = u + at \rightarrow v = 9.8 \times 2 = 19.6$	M1 A1 (2)
(c)	Stage 1 distance $\frac{1}{2} \times 9.8 \times 4$ OR $\frac{1}{2} \times 2 \times 19.6 = 19.6$ Stage 2 distance $\frac{1}{2} (19.6 + 4) \times 5$ (or equivalent two stage method) $= 59$ (acceleration = 3.12 M1 A1, 59 A1) Minimum height for h = 59 + 19.6 = 78.6 m	B1 ft M1 A1 ft A1 cas A1 f.t. (5)
(d)	From a height of 125m, there are 46.4m to fall at 4 ms⁻¹ Time for stage 3 = $\frac{46.4}{4} \text{ s} \rightarrow (11.6 \text{ s})$ Total time = 2 + 5 + 11.6 $\rightarrow 18.6 \text{ s}$	M1 M1 A1 ft A1 cas (4)
(e)	Air resistance in (a) or equivalent sound reason	B1 (1)

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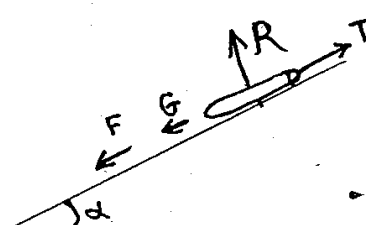
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7. (a)	$\alpha = \arctan \frac{5}{12}$ $\cos \alpha = \frac{12}{13}$, $\sin \alpha = \frac{5}{13}$ $R = 78g \cos \alpha$ $F = 78g \cos \alpha (0.25)$ $G = 78g \sin \alpha$  Newton II along slope attempted with T, F, G included $T - F - G = 78(0.5)$ Solving for T (dependent on M1) $T = 509.4$ (accept this or 510 to 2 s.f. or 509 to 3 s.f. result only)	M1 A1 B1 M1 A1 f.t. B1 M1 M1 A1 (9)
(b)	Accelerating force down slope is $G - F$ (or Friction reversed and T no longer included) Newton II $G - F = 78a$ $a = g \sin \alpha - \mu g \cos \alpha$ $= 9.8 \left(\frac{5}{13} - \frac{3}{13} \right)$ or $\frac{2}{13}g$ $= 1.5, 1.50, 1.51$ // Score A2 other answers which round to 1.5 Score A1	M1 M1 A1 M1 A2, 1.5 (6)

H.M.K.
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